Health Care FMEA

EHR Drug-Allergy Alert Verification & Override Governance Process

Failure Mode and Effects Analysis (FMEA)_Five Process Steps

Process Step #1 Allergy Data Entry and Verification at Admission	
Box 1 **Process Step**	Allergy Data Entry and Verification at Admission
Box 2 **Potential Failure Mode**	Patient allergy information is entered incompletely, inaccurately, or not entered at all into the EHR at the time of admission.
Box 3 **Potential Cause(s)**	• Nursing staff rely on verbal self-report without cross-referencing prior medical records or pharmacy records. • EHR admission workflow does not enforce a mandatory allergy-documentation field before proceeding. • High patient volume and time pressure lead to shortcuts in data entry. • Inconsistent use of allergy terminology (free-text vs. coded entries) causes structured alert logic to fail.
Box 4 **Severity**	Catastrophic (10) – An unrecorded allergy can result in a fatal anaphylactic reaction or permanent physiologic injury.
Box 5 **Probability**	Occasional – Incomplete allergy entry at admission occurs several times per 1–2 years in busy inpatient units.
Box 6 **Hazard Score**	12 (Catastrophic × Occasional)
Box 7 **Action (Eliminate, Control, or Accept)**	Eliminate
Box 8 **Description of Action**	• Implement a mandatory, system-enforced allergy verification step in the EHR admission workflow; the record cannot be finalized until at least one allergy status is documented (NKDA, or specific allergen with reaction type). • Integrate pharmacy benefit manager (PBM) and prior-record data feeds to auto-populate known allergies for clinician review and confirmation. • Require dual-source verification for all high-risk allergens (penicillin, sulfonamides, NSAIDs, contrast media): admitting nurse documents, and pharmacist reconciles within two hours of admission.

Process Step #2 Clinical Decision Support (CDS) Allergy Alert Firing	
Box 1 **Process Step**	Clinical Decision Support (CDS) Allergy Alert Firing
Box 2 **Potential Failure**	The CDS allergy alert fails to fire, fires incorrectly, or is suppressed before the ordering clinician sees it.

Mode	
Box 3 Potential Cause(s)	• Alert is configured at a severity threshold that does not trigger for cross-reactive drug classes. • Alert fatigue causes the system to auto-suppress or soft-fire the alert, bypassing the hard-stop workflow. • Incomplete formulary mapping means a brand-name order does not match the generic allergen record. • System downtime or interface failure at the point-of-care workstation prevents alert delivery.
Box 4 Severity	Catastrophic (10) – Failure of the alert to fire can directly lead to a fatal adverse drug event.
Box 5 Probability	Uncommon – With proper CDS configuration, technical alert failures may occur once in 2–5 years, but are clinically catastrophic when they do.
Box 6 Hazard Score	8 (Catastrophic × Uncommon)
Box 7 Action (Eliminate, Control, or Accept)	Eliminate
Box 8 Description of Action	• Configure allergy alerts as interruptive hard-stops for all Severity Class I and II drug-allergen interactions, requiring an explicit override with reason code rather than passive acknowledgment. • Conduct quarterly CDS logic audits comparing the formulary drug-allergen mapping table against updated clinical databases (e.g., First Databank, Multum) to catch gaps in cross-reactive class coverage. • Establish a redundant alerting pathway: if a workstation alert is not acknowledged within 60 seconds, a secondary alert is pushed to the ordering clinician's mobile device and the unit charge nurse.

Process Step #3 Clinician Override of Allergy Alert	
Box 1 Process Step	Clinician Override of Allergy Alert
Box 2 Potential Failure Mode	A clinician overrides a high-severity allergy alert without clinical justification, or the override is granted without appropriate secondary review.
Box 3 Potential Cause(s)	• Override reason codes are too generic (e.g., "Benefit outweighs risk") and require no clinical narrative. • No pharmacist co-sign requirement exists for overrides of Severity Class I alerts. • Alert fatigue from high override rates normalizes the behavior; peer norms discourage pushback. • Lack of real-time audit visibility means supervisory staff are unaware of override patterns.
Box 4	Catastrophic (10) – An unjustified override is the direct mechanism by which a fatal allergic reaction occurs.

Severity	
Box 5 **Probability**	Occasional – Without a mandatory co-sign process, unjustified overrides of high-severity alerts likely occur several times per year.
Box 6 **Hazard Score**	12 (Catastrophic × Occasional)
Box 7 **Action (Eliminate, Control, or Accept)**	Eliminate
Box 8 **Description of Action**	• Require mandatory pharmacist co-signature for any override of a Severity Class I allergy alert before the order can be dispensed; no medication leaves the pharmacy until the co-sign is electronically recorded. • Replace vague override reason codes with structured clinical justification fields (drug class necessity, allergy date/validity, premedication protocol attached); free-text override reasons must be a minimum of 25 characters. • Generate a same-day automated alert to the patient safety officer and unit medical director for every Severity Class I override, and include monthly aggregate override reports in departmental quality dashboards.

Process Step #4 Pharmacist Dispensing Review Prior to Medication Release	
Box 1 **Process Step**	Pharmacist Dispensing Review Prior to Medication Release
Box 2 **Potential Failure Mode**	The pharmacist verifies and dispenses the medication without detecting or acting on the allergy conflict.
Box 3 **Potential Cause(s)**	• Pharmacist's verification screen does not prominently display allergy warnings alongside the order being reviewed. • High-volume dispensing workflow creates time pressure; pharmacists skim verification screens. • Pharmacist lacks access to the full override reason documented by the prescriber. • Shift handoff results in a different pharmacist verifying the order without context from earlier clinical discussions.
Box 4 **Severity**	Major (7) – Dispensing an allergen-contraindicated drug without detection allows a potentially fatal dose to reach the patient; severity is Major because this is a last-line catch step, not the initiating failure.
Box 5 **Probability**	Occasional – In environments without structured pharmacist allergy alerts, dispensing errors recur within 1–2 year cycles.
Box 6 **Hazard Score**	9 (Major × Occasional)
Box 7 **Action (Eliminate,**	Control

Control, or Accept)	
Box 8 Description of Action	• Redesign the pharmacist's order-verification screen to display a persistent, color-coded allergy conflict banner (red for Severity Class I) that cannot be scrolled past or collapsed during the active verification session. • Require the pharmacist to document a clinical verification note—separate from the prescriber's override—confirming review of the allergy conflict and the rationale for release; this note becomes a permanent part of the medication administration record (MAR). • Implement a mandatory pharmacy-to-prescriber callback protocol for all Severity Class I override orders: the pharmacist must verbally confirm clinical intent with the ordering provider and log the confirmation before dispensing.

Process Step #5 Nurse Administration Verification and Patient Identification Check	
Box 1 Process Step	Nurse Administration Verification and Patient Identification Check
Box 2 Potential Failure Mode	The nurse administers the allergen-conflicted medication to the patient without identifying or escalating the allergy concern.
Box 3 Potential Cause(s)	• The Five Rights check is performed without reviewing the patient's allergy band or EHR allergy profile. • Bedside medication administration record (eMAR) does not present a visible allergy alert at the time of scanning. • Nurse is not aware of the prescriber's override rationale and assumes the allergy concern was already resolved upstream. • Patient is unable to report their own allergy (sedated, pediatric, or cognitively impaired) and no surrogate is present to verify.
Box 4 Severity	Catastrophic (10) – Administration is the final point of intervention; failure here results directly in patient exposure to a known allergen, with potential for anaphylaxis and death.
Box 5 Probability	Uncommon – With upstream controls functioning, the probability of nursing-level failure in isolation is reduced; however, when upstream failures occur together, nursing is the last safeguard.
Box 6 Hazard Score	8 (Catastrophic × Uncommon)
Box 7 Action (Eliminate, Control, or Accept)	Control
Box 8 Description of Action	• Configure the eMAR barcode-scanning workflow to trigger an interruptive allergy alert pop-up at the bedside terminal when a Severity Class I allergen conflict is present, requiring the nurse to enter their employee ID and a reason code before the scan is accepted as complete. • Establish a mandatory verbal allergy confirmation protocol as part of the pre-administration patient identification check: the nurse asks the patient (or

	surrogate) to state their known allergies before any high-risk medication is administered, and documents the patient’s response in the eMAR. • Conduct bi-annual simulation training for nursing staff using allergy-alert scenario exercises, with competency validation required; track completion rates and debrief any real-world allergy near-miss events within 72 hours using a structured learning-from-defects framework.